

Influence of muscle strength, power and rapid force capacity on maximal club head speed in male National Level golfers

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1 **INFLUENCE OF MUSCLE STRENGTH, POWER, AND RAPID FORCE**
2 **CAPACITY ON MAXIMAL CLUB HEAD SPEED IN MALE NATIONAL**
3 **LEVEL GOLFERS**

4
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9
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11 force development

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16 **ABSTRACT**

17 This study investigated relationships between maximal club head speed (CHS) and physiological
18 and anthropometric parameters in twenty-one national level male golfers (age: 21.9 ± 3.9 years;
19 handicap: $+1.1 \pm 1.7$).

20 Maximal isometric strength (MVC) was measured during isometric mid-thigh pull and bench
21 press, while MVC and rate of force development (RFD) were measured during isometric leg press.
22 Power, lower limb stiffness, positive impulse, jump height and RFD_{dyn} were measured during
23 countermovement jump (CMJ). Moreover, rotational trunk power, active range of motion (AROM)
24 and anthropometrics were determined. Comparisons were made between participants with high
25 (FTG) and lower (STG) CHS respectively.

26 FTG demonstrated greater isometric mid-thigh pull and isometric bench press MVC, leg press
27 RFD, rotational trunk power, and CMJ parameters (except RFD_{dyn}) as well as reduced hip AROM
28 compared to STG ($P < 0.01$). CHS was positively correlated to isometric mid-thigh pull and
29 isometric bench press MVC, leg press RFD, rotational trunk power and CMJ parameters ($P < 0.01$).

30 In conclusion, strong positive correlations were observed between maximal CHS and maximal
31 strength and power parameters. Consequently, improving maximal neuromuscular strength and
32 power may be considered of importance for golfers, as greater CHS and accompanying driving
33 distance may lead to competitive advantages.

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INTRODUCTION

Performance in competitive golf is determined by the ability to complete a tournament with as few strokes as possible. Accordingly, golfers are required to master a variety of specific skills such as driving, approach play, short game and putting as well as strategic and psychological elements. The technical, technological, psychological, and tactical elements have traditionally been considered the primary performance determinants (Farraly et al. 2003; Whittaker A. 1998), while recent studies have also emphasized the importance of maximal driving distance for golf performance. Hellström et al. (2014) observed negative correlations between stroke distance and golf score on par 4 ($r = -0.45$, $P < 0.001$) and par 5 ($r = -0.54$, $P < 0.001$) holes, indicating that longer driving distances are associated with lower scores. Moreover, professional golfers are likely to gain 0.75 strokes per round on their opponents by driving 20 yards further (Broadie. 2014).

Driving distance relies on several factors including equipment properties (Kenny et al. 2008, Worobets and Stefanyshyn 2012), swing technique (e.g., grip positioning, hand path length, x-factor and x-factor stretch, kinematic sequence) (D'Arcy et al. 2021; Mackenzie et al. 2020; Nesbit and Serrano; 2005), centeredness of strike (Betzler et al. 2014; Sweeney et al. 2013), ball temperature (Carlsson et al. 2019) and environmental factors (Cochran and Stobbs, 1999).

While driving distance and driving accuracy are factors direct transferable to golf performance, the aforementioned variables can have significant confounding influence on these measurements. Hence, club head speed at ball impact (CHS) has been widely used as an independent parameter reflecting golf performance in the literature (Coughlan et al. 2020; Keogh et al. 2009; Leary et al. 2012; Sheehan et al. 2019; Wells et al. 2018; Wells et al. 2019; Wells et al. 2022). High CHS has been reported by multiple studies to be decisive for maximal ball speed, carry distance and driving distance (Donahue et al. 2021; Fletcher and Hartwell. 2004; Sweeney et al. 2013) and to be related to overall golf performance (i.e., golf handicap (HCP)) (Fradkin et al. 2004). Additionally, CHS is a key focus point of attention for strength and conditioning practitioners and scientists in golf since it can be improved through strength and conditioning exercise (Ehlert 2020; Torres Ronda et al. 2011; Uthoff et al. 2021)

Emphasizing the importance of physical training in golf, various training modalities (strength training, flexibility training, plyometric exercises, or mixed-type training efforts) have been shown to induce positive effects on CHS and golf performance in different populations (player levels) (Alvarez et al. 2012; Cummings et al. 2018; Doan et al., 2006; Fletcher and Hartwell, 2004; Hetu et al. 1998; Lephart et al., 2007; Oranchuk et al. 2018; Thompson et al., 2007; Thompson and Osness,

2004).

The relative activation of individual muscles and muscle groups in the golf swing have been identified, more specifically, the pectoralis major, external and internal obliques and erector spinae are all important contributors to assist sequential rotation of the torso and upper limbs to hereby transfer force to the golf ball (Cole and Grimshaw, 2016; McHardy and Pollard, 2005). In the lower body the biceps femoris, gluteus maximus, gluteus medius, adductor magnus and vastus lateralis similarly provide support for force transfer and rotation of the upper body (Cole and Grimshaw, 2016; McHardy and Pollard, 2005). Moreover, Nesbit and Serrano (2005) observed the lumbar spine ($\approx 24\%$), right hip ($\approx 19\%$) and the thoracic spine ($\approx 18.5\%$) as the three body regions that performed the most total work during the golf swing in a sample containing four mixed level right-handed golfers. Hence, improving neuromuscular function (i.e., maximal strength, power, rate of force development (RFD), and flexibility) of these muscle groups may allow golfers to perform more mechanical work on the golf club during the swing, thus creating higher CHS.

High contractile forces are produced during the golf swing (Hume et al. 2005; McHardy and Pollard, 2005). Consequently, maximal muscle strength has been positively linked with CHS and golf performance (Ehlert 2021; Smith et al. 2011; Torres-Ronda et al. 2011). Thus, Keogh et al. (2009) reported a very strong ($r = 0.71$) correlation between trunk rotator strength assessed using a golf swing specific cable woodchop test and maximal CHS as well as a tendency for hack squat ($r = 0.53$) to also be positively related to CHS in mixed level male golfers. Similarly, Hellström et al. (2008) demonstrated a strong correlation between 1 repetition maximum (RM) squat ($r = 0.54$) and CHS in elite (HCP +5 to 0) male golfers. Moreover, peak force recorded during isometric mid-thigh pull has been shown to be associated with CHS in highly skilled ($r = 0.48$) and recreational ($r = 0.2$) male golfers (Leary et al. 2012; Wells et al. 2018) and very strongly correlated to driver carry distance in high level youth golfers (Sanders et al. 2020). Moreover, Wells et al (2022) observed a strong correlation ($r = 0.52$) between CHS in highly skilled male golfers and countermovement jump (CMJ) peak force. Also, upper body dynamic strength assessments (1-6 RM bench press) have revealed strong correlations between strength and maximal CHS or ball speed in mixed level male golfers and highly skilled young male golfers (Keogh et al. 2009; Torres-Ronda et al. 2014). Furthermore, Ehlert (2021) reported maximal CHS to be moderately associated with upper and lower body muscle strength ($r = \approx 0.43$) with maximal 1RM squat strength specifically being strongly correlated to CHS ($r = 0.63$). Moreover, Wells et al (2022) observed a strong correlation between CHS in highly skilled male golfers and countermovement jump peak force.

102 The golf downswing represents the strike phase (duration < 300 ms) where the club is accelerated
103 towards the ball using a maximum of muscular activity (Hume et al., 2005; McHardy and Pollard,
104 2005). Consequently, it is suggested that muscle power and RFD are important parameters for
105 improving CHS (Read et al. 2014; Sheehan et al., 2019). Numerous tests of lower body power in
106 golfers have revealed significant relationships between CHS and golf performance. Thus, CMJ
107 jump height ($r = 0.18 - 0.72$) (Leary et al. 2012; Oranchuk et al. 2018; Read et al. 2013; Wells et al.
108 2009; Wells et al. 2022) peak power ($r = 0.22 - 0.66$) (Hellström 2008; Leary et al. 2012; Wells et
109 al. 2022) and positive impulse ($r = \approx 0.75$) (Wells et al. 2018; Wells et al. 2022) have shown
110 correlations ranging from weak to very strong with CHS or ball speed in different populations.
111 Studies investigating the influence of RFD on CHS are limited in the literature, Wells et al. (2018)
112 assessed RFD in highly skilled golfers during the isometric mid-thigh pull and observed moderate
113 correlations between CHS and isometric mid-thigh pull RFD at 0-150 and 0-200 ms, while Sanders
114 et al. (2020) found strong relationships between driver CHS and isometric mid-thigh pull force at
115 250 ms and isometric mid-thigh pull RFD at 0-250 ms in high level youth golfers. Moreover, Leary
116 et al. (2012) found isometric mid-thigh pull force at 0–150 ms to be moderately correlated with
117 CHS in recreational golfers. For the assessment of relationship between upper body muscle or
118 rotational trunk power and CHS, the most common test evaluated is standing rotational medicine
119 ball throw with weak to very strong correlations ($r = 0.29 - 0.71$) when compared to CHS or drive
120 ball speed in different populations (Coughlan et al. 2020; Donahue et al. 2021; Gordon et al. 2009;
121 Read et al. 2013; Torres-Ronda et al. 2014).

122 Joint flexibility may have positive impact on force production in the golf swing as a greater
123 range of motion may allow for a longer backswing which would allow for more time to generate
124 force and perform greater levels of work during the downswing (Keogh et al., 2009). However,
125 mixed results have been reported as some studies have found no significant relationship between
126 CHS and trunk, hip, and wrist flexibility (Gordon et al. 2009; Keogh et al. 2009). Whereas other
127 have demonstrated larger range of motion (ROM) to be beneficial for golf performance (Doan et al.
128 2006; Hetu et al. 1998; Sell et al. 2007). Finally, a recent meta-analysis showed no significant
129 correlation between rotational trunk flexibility ($r = 0.17$) sit and reach flexibility ($r = 0.08$) or
130 general joint flexibility ($r = 0.03$) versus CHS (Ehlert, 2021).

131 Inconsistent findings have been reported for the relationship between anthropometry and golf
132 performance incl. CHS. Keogh et al. (2009) found no significant correlations between
133 anthropometric measures and golf performance, although reporting a tendency for golfers with low

HCP to have small to moderately greater arm length, fat-free mass and larger chest and arm girth measurements than golfers with higher HCP. Additionally, Donahue et al. (2021) found a weak correlation ($r = 0.29$) between body mass and CHS while finding a strong correlation ($r = 0.55$) between body height and CHS in male collegiate golfers. Other studies have shown moderate to very strong correlations between arm length ($r = 0.71$), body height ($r = 0.71$) and driving carry distance in top amateur golfers (Wells et al. 2009) and between body mass and CHS in youth male and female golfers ($r > 0.4$) (Coughlan et al. 2020) and elite (HCP +5 to 0) male golfers ($r = 0.51$) (Hellström et al. 2008). Finally, Ehlert (2021) reported moderate correlations between body height ($r = 0.31$) and body mass ($r = 0.44$) and CHS in a meta-analysis, overall indicating a tendency for CHS to be positively correlated to body height and body mass.

Notably, a majority of the above-mentioned studies have included recreational or less skilled golfers or have focused more narrowly on specific parameters such as lower body strength and power. Less skilled golfers present larger variability in CHS (Betzler et al., 2012), hence, establishing relationships between physical capacity and CHS can be confounded by a golfer's ability (swing technique). Therefore, to expand the understanding of optimizing high level golf performance, it is highly relevant to identify specific physical characteristics that underlie CHS and driving distance performance in highly skilled golfers.

The primary aim of this study, therefore, was to examine the relationship between CHS and selected physical characteristics in highly skilled golfers, including anthropometry, active range of joint motion (AROM), upper- and lower body strength and contractile power, as well as explosive force capacity. Additionally, it was examined how these physical characteristics differed between highly skilled male golfers with higher vs. lower CHS performance.

Based on previous literature, we hypothesized that upper- and lower body strength, contractile power, and RFD would be positively correlated with CHS in the present population of highly skilled male golfers. Further, we hypothesized that anthropometric characteristics (arm length, body height, body mass) and AROM would correlate positively with maximal CHS.

160

MATERIAL & METHODS

Participants

Twenty-one national level (also termed *highly skilled*) (McKay et al. 2022) right-handed male golfers (age: 21.9 ± 3.9 years, height: 184.6 ± 4.4 cm, body weight: 80.6 ± 11.9 kg, mean arm

length: 75.0 ± 2.7 cm, HCP: $+1.1 \pm 1.7$) were included in the study (Table 1) that included a familiarization session (14 days prior) and an actual 2-hour test session. Inclusion criteria were HCP ≤ -2 , age > 18 years of age and injury free. The sample included 13 golfers with “professional” status competing at the domestic Access Tour, with all participating participants competing at the national elite level. All participants had well established experience with resistance training and physical performance testing. Participants were informed of the methods and the potential risks and benefits of this study and gave informed approval prior to participation. The study was conducted with established ethical standards, approved by the local ethics committee. Participants were instructed to be well hydrated upon arrival in the laboratory, to maintain their normal diet and to refrain from any form of excessive physical exercise 48 hours prior. To enable comparison between players with fast club head speed vs. less fast club head speed, the cohort was divided *post hoc* into top and bottom tertiles based on their CHS, with FTG ($n = 7$) representing the tertile of players with highest CHS and STG ($n = 7$) representing the tertile of players with lowest CHS. The mid-tertile ($n = 7$) representing the average CHS of the entire player cohort was not included in this sub-analysis.

[Insert Table 1 approximately here]

Study overview

A cross-sectional design including a between group comparison was employed to investigate the relation between CHS and selected physical variables of contractile muscle performance and function. All participants underwent a standardized test protocol, with tests performed in the following order: Anthropometric measures, CHS measures, CMJ, isometric leg press, isometric mid-thigh pull, isometric bench press, cable trunk rotation power, AROM. All tests were separated by five minutes.

Anthropometrics

All measurements were conducted with the participants wearing shorts, t-shirt, and no shoes. Body weight was measured to the nearest 0.1 kg using a digital scale (Soehnle, Backnang, Germany). Subsequently, body height was measured to the nearest 0.5 cm with a stadiometer Model 213 (SECA, Hamburg, Germany). With the participant standing against a wall (90° shoulder flexion, full elbow extension and pronation of the forearm), the distance from the lateral edge of the acromion to the most distal aspect of the third finger was noted (accuracy 0.5 cm) as the arm length. The mean length of the left and right arm was used for further analysis.

195 *Club head speed*

196 CHS was recorded during repeated maximal golf drives using radar-based equipment (TrackMan 4
197 launch monitor system, Vedbæk, Denmark) (Coughlan et al. 2020, Wells et al. 2019), a special
198 made receiver net and an indoor golf mat. The system settings were based on guidelines from the
199 manufacturer, which included the setting 'indoor mode' and the assessor specifying the intended
200 target line. The participants used their own custom fitted driver and unused Srixon Z-star (2020)
201 premium golf balls. Prior to testing, all participants conducted a standardized warm up session
202 consisting of 5-minutes warm-up on a bicycle ergometer against a standard load (60 W) followed
203 by a series of dynamic upper and lower body mobilization movements (wall slides, page turns, arm
204 swings, leg crossovers, leg swings, clocklunges, overhead squats), light upper body bands pull and
205 push exercises, dynamic glute activation (fire hydrants and gluteal bridges), 3-5 plyometric jumps
206 and a self-selected number of warm-up shots with progressively increasing CHS (Sanders et al.,
207 2020; Wells et al., 2018). Subsequently the participants were instructed to aim down the target line,
208 and to strike the ball using maximum effort while maintaining normal swing mechanics (Coughlan
209 et al. 2020, Wells et al. 2019). Each participant performed five drives from a self-selected tee-height
210 separated by at least 30 seconds. If the fifth drive showed the highest CHS, additional drive shots
211 (with a limit of 2) were performed until a decline in CHS was observed. The greatest speed
212 achieved was noted as individual CHS.

213 *Countermovement jumping*

214 Following 3-5 submaximal attempts, the participants performed five maximal CMJ on a force plate
215 (AMTI R6-5-1000, Watertown, MA) with analog ground reaction force (F_z) signals recorded at
216 1000 Hz using an external A/D converter (DT 2801A, Data Translation) (Bojsen-Moller et al.,
217 2005; Jakobsen et al., 2012). From a standing position (arms akimbo), the participants moved
218 rapidly downwards to individually preferred squat-depth, immediately followed by a fast upward
219 movement. The participants were instructed to “jump as high as possible”. (Bojsen-Moller et al.
220 2005, Caserotti et al. 2001, Wells et al. 2019). All jumps were separated by at least 90 seconds. The
221 attempt with the greatest individual jump height was chosen for further analysis.

222 The CMJ *post hoc* analysis was conducted as previously described (Bojsen-Moller et al., 2005;
223 Caserotti et al., 2001; Holsgaard Larsen et al., 2007; Jakobsen et al., 2012; Thorlund et al., 2008). In
224 brief, the vertical velocity (V) of the body center of mass (BCM) was derived by time integration of
225 the ground reaction force (F_z) signal converted into acceleration ($F_z/m - g$, where m = body mass
226 (kg), $g = 9.81 \text{ ms}^{-2}$). BCM-position was obtained by subsequent time integration of V . Outcome

variables were: Maximal jump height (Bojsen-Møller et al. 2005), lower limb stiffness (Jakobsen et al. 2012; Jordan et al. 2018; Jorgensen et al. 2021), dynamic rate of force development (RFD_{dyn}) at 0-50 ms (Aagaard et al. 2002; Jakobsen et al. 2012; Thorlund et al., 2008), positive impulse (Jordan et al. 2018; Wells et al. 2018) and peak and mean power generated during the concentric take-off phase (termed respectively CMJ peak power and CMJ mean power) (Holsgaard Larsen et al. 2007) (Holsgaard Larsen et al. 2007). These parameters (CMJ jump height, impulse, peak and mean power) are previously reported to be highly reliable (Holsgaard Larsen et al. 2007; Merrigan et al. 2021; Souza et al. 2020; McLellan et al. 2011). Further, moderate-to-high intersession reliability has been demonstrated for lower limb stiffness obtained during CMJ testing (CV 5.1–9.3%, ICC 0.78–0.95) (Moir et al. 2009), while CMJ RFD_{dyn} have shown low-to-moderate test-retest reliability (Merrigan et al. 2021; McLellan et al. 2011; Souza et al. 2020).

238

239 *Isometric leg press strength and RFD*

Participants were seated in a rigid custom-made leg press bench with both feet on a force plate (Kistler 9367/8 B, Switzerland) and with hip and knee angles at approximately 100° and 70° (*in situ* joint position 0°), respectively (Marcora et al. 2000, Zaras et al. 2016, Zaras et al. 2022). The isometric leg press test is reported to be highly reliable when measuring peak force (Martinopoulou et al. 2022), while moderate-to-high test-retest reliability has previously been reported for the assessment of RFD in isometric leg press (Bojsen-Møller et al. 2005, Buckthorpe et al. 2012, Maffiuletti et al. 2007, Martinopoulou et al. 2022). Three warm-up efforts were conducted at approximately 50, 75 and 90% of perceived maximal intensity. Three maximal trials were then performed separated by at least two minutes rest. Participants were instructed to exert force as fast and hard as possible and received strong verbal encouragement during all efforts. Orthogonal leg press force (F_z data) was recorded at 1000 Hz using an external A/D converter (16-bit National Instrument USB-6002 card), and individual peak force was noted (instantly displayed following contraction) as the greatest of the three trials.

Contractile RFD was measured in the trial with the highest resultant peak force (Aagaard et al., 2002, Caserotti et al. 2008). RFD ($\Delta F_z / \Delta \text{time}$) was calculated as the mean tangential slope of the force–time curve in the initial 30, 50, 100 and 200 ms of contraction. Onset of force was defined as the instant (time index) when leg press force increased 2% above the baseline force (2% of the peak MVC force measured within the same sweep) (Granacher et al. 2009, Maffiuletti et al. 2016).

258 *Isometric mid-thigh pull and isometric bench press strength*

259 Isometric mid-thigh pull testing was conducted using a commercially available device (Kairos
260 Strength, USA). The isometric mid-thigh pull is shown to be reliable when measuring for peak
261 force both within (ICC = 0.98 and 0.97) and between sessions (ICC = 0.89) (De Witt et al. 2018).
262 A force plate (AMTI R6-5-1000, Watertown, MA) was placed inside the device, and a weightlifting
263 bar (Eleiko) was fixed at a specific target height relative to the platform. Comfort et al. (2015)
264 suggests performing the isometric mid-thigh pull at individually preferred knee and hip angles
265 while keeping the bar resting at the midpoint (50%) of the thigh. This individual adjustment of
266 posture is likely to result reduced learning effects and high test-retest reliability (Comfort et al.,
267 2015). Across the present participants, this resulted in an average knee angle of around 125° and
268 average hip angle of around 145° (full standing extension = 180°). Furthermore, according to Guppy
269 et al. (2018) maximal isometric mid-thigh pull force output appears to remain unaffected by
270 alterations in body posture provided the barbell is placed at the midpoint (50%) of the thigh. Hence,
271 participants were instructed to stand on the force plate with feet directly below the bar (hip width
272 apart). The bar was set at the midpoint of the thigh (50%) and the participants took a position with
273 an upright torso, self-selecting the hip and knee-joint angles and then the bar was fixed in this
274 position. During the isometric mid-thigh pull attempts participants hands were strapped firmly to
275 the bar by means of weightlifting (wrist) straps. Three submaximal pulls at approximately 50, 75
276 and 90% of maximal effort served as specific warm-up. Subsequently, each participant performed
277 three maximal trials with the instruction to pull the bar and push against the ground as 'hard and as
278 fast as possible' for five seconds while verbal encouragement was provided (Haff et al. 2015, Wells
279 et al. 2019). Prior to each maximal trial, the participants were instructed to position themselves
280 while using some pre-tension to minimize slack in the body and equipment. Each trial was
281 separated by two minutes and the trial with the greatest vertical peak force (N) was used for further
282 analysis.

283 Isometric bench-press MVC testing was performed using the same experimental setup as
284 described for the isometric mid-thigh pull test. The isometric bench press test has been reported to
285 be highly reliable when performed by elite athletes when measuring peak force (ICC = 0.89 to 0.97,
286 %CV = 1.2 to 1.6) (Young et al. 2014). Participants were positioned on the force plate lying prone
287 on their back (entire body weight supported by the force plate), and with their knees and hips flexed
288 to ensure no ground contact outside the force plate. The barbell was positioned above the sternum at
289 a height of 50% of arm length relative to the force plate, such that the elbow angle was

290 approximately 70-80° (*in situ* position = 0°) (Loturco et al. 2016). The warm-up and test protocols
291 were identical to that described for the isometric mid-thigh pull. The trial with the highest PF (N)
292 was selected for further analysis.

293

294 *Cable trunk rotation power*

295 Trunk rotation power was assessed using an experimental set-up previously employed in golfers
296 (Keogh et al. 2009, Schofield et al. 2021). The test is previously shown to be reliable (CV < 10%,
297 ICC ≥ 0.90 with test loads ranging from 11.25 kg to 18.75 kg) when measuring vertical cable stack
298 velocity (Schofield et al. 2021). In brief, a linear distance encoder (displacement transducer, 1000
299 Hz) (T-Force (dynamic measurement system, ERGOTECH Consulting, Murcia, Spain)) was
300 calibrated according to manufacturer guidelines and fixed to the weight stack of a strength training
301 cable device. The cable was fixed to a custom-made handle that resembled a golf club. Participants
302 were standing 1 m anteriorly to the cable rack, and with a load of 15 kg, the participants were
303 instructed to mimic their golf downswing starting from a six-iron backswing posture while
304 producing maximal force throughout the impact phase and stopping in a position with the left arm
305 elevated (Schofield et al. 2021). Warm-up and testing were conducted similarly to that described for
306 isometric mid-thigh pull. Peak power and mean power were recorded and the trial with the highest
307 peak power (W) was selected for further analysis. If peak power increased throughout all three
308 attempts, a fourth attempt was allowed. Trials were rejected if participants showed a substantial loss
309 of starting posture or could not control the eccentric motion after execution of the downswing
310 (Schofield et al. 2021).

311 *Active range of joint motion (AROM)*

312 Bilateral measurements were performed using an analog goniometer while participants were
313 actively displacing the relevant joint: Maximal shoulder internal and external rotation, trunk
314 rotation and hip internal and external rotation was recorded as previously reported (Marshall and
315 Llewellyn. 2017, Sheehan et al. 2019). Three measurements were obtained for each joint. The
316 average value was calculated and summed for each joint to reflect total AROM (Marshall and
317 Llewellyn. 2017).

318 *Statistical analysis*

319 Group mean data (+/- SD) were calculated and reported below for all obtained outcome variables.
320 Linear regression analysis (Pearson product-moment correlation) was performed between CHS and
321 all relevant outcome variables. Strength of correlations was defined as weak: $r < 0.30$, moderate: $r =$

0.30 to 0.49, strong: $r = 0.50$ to 0.69 , very strong: $r = 0.70$ to 0.89 , nearly perfect: $r = 0.90$ to 0.99 , or perfect: $r = 1.0$ (Hopkins 2012, Leary et al. 2012). Student's unpaired t-tests (alpha level of $P \leq 0.05$, two-tailed) were applied to determine differences between FTG and STG players.

Intra-session variability was assessed by calculation of within-subject coefficient of variation (Holsgaard et al. 2007, Vasikaran et al. 1994).

327

328 RESULTS

329 *Golf performance*

330 A strong positive correlation was observed between CHS and HCP ($r = 0.59$, $r^2 = 0.35$, $P = 0.005$).
331 Furthermore, FTG demonstrated 11.3% greater CHS ($P \leq 0.0001$) and better HCP ($P = 0.04$)
332 compared with STG.

333 *Anthropometric measures*

334 No associations were observed between CHS and any of the anthropometric measures. Moreover,
335 no between-group differences were observed for any of the anthropometric parameters.

336

337 [Insert Figure 1 and Table 2 approximately here]

338

339 *Maximal muscle strength and power*

340 Very strong positive correlations were observed between CHS and isometric mid-thigh pull peak
341 force ($r = 0.82$, $r^2 = 0.67$, $P \leq 0.0001$), isometric bench press peak force ($r = 0.76$, $r^2 = 0.58$, $P \leq$
342 0.0001), cable trunk rotation peak power ($r = 0.81$, $r^2 = 0.65$, $P \leq 0.0001$) (Figure 1) and isometric
343 leg press RFD 200 ms ($r = 0.70$, $r^2 = 0.49$, $P = 0.002$) (Table 2). Furthermore, strong positive
344 correlations were observed between CHS, and cable trunk rotation mean power ($r = 0.53$, $r^2 = 0.28$,
345 $P = 0.01$) and isometric leg press RFD at 30 ms ($r = 0.59$, $r^2 = 0.35$, $P = 0.01$), 50 ms ($r = 0.63$, $r^2 =$
346 0.4 , $P = 0.006$), and 100 ms ($r = 0.65$, $r^2 = 0.42$, $P = 0.005$). Between group differences (FTG >
347 STG) were observed for isometric mid-thigh pull peak force (+48%, $P = 0.0007$) and isometric
348 bench press peak force (+36%, $P = 0.014$) (Figure 2). Isometric leg press peak force tended to be
349 greater for FTG compared to STG (+19%, $P = 0.09$). Isometric leg press RFD at 30 ms (+104%, P
350 $= 0.044$), 50 ms (+116%, $P = 0.04$), 100 ms (+114%, $P = 0.02$) and 200 ms (+83%, $P = 0.009$) were

all greater in FTG than STG. Finally, cable trunk rotation peak power (+37%, $P = 0.0001$) and mean power (18%, $P = 0.018$) were greater in FTG than STG (Figure 2).

Countermovement jumping

Very strong relationships were observed between CHS and CMJ jump height ($r = 0.75$, $r^2 = 0.56$, $P \leq 0.0001$), CMJ positive impulse ($r = 0.76$, $r^2 = 0.57$, $P \leq 0.0001$), CMJ mean power ($r = 0.81$, $r^2 = 0.66$, $P \leq 0.0001$), CMJ peak power ($r = 0.73$, $r^2 = 0.54$, $P \leq 0.0001$) and CMJ lower limb stiffness ($r = 0.77$, $r^2 = 0.60$, $P \leq 0.0001$). Moreover, a strong positive correlation was observed between CHS and CMJ RFD_{dyn} ($r = 0.61$, $r^2 = 0.37$, $P = 0.004$) (Table 2).

FTG performed better than STG in terms of maximal vertical CMJ jump height (+28%, $P = 0.004$), CMJ positive impulse (+26%, $P = 0.002$), CMJ peak power (+25%, $P = 0.008$), CMJ mean power (+33%, $P = 0.006$) and CMJ lower limb stiffness (+53%, $P = 0.04$) (Table 3). No significant difference was observed between the groups for CMJ RFD_{dyn}, although FTG tended statistically to have higher RFD_{dyn} than STG (+69%, $P = 0.11$).

Active range of joint motion:

No relationships were observed between CHS and AROM measures (Table 2). However, CHS tended to be negatively related to total AROM of the right hip ($r = -0.42$, $r^2 = 0.17$, $P = 0.06$) and STG had significantly greater AROM in the left hip (+17%, $P = 0.01$) and right hip (+16%, $P = 0.01$) compared to FTG (Table 3).

Intra-session variability

Intra-session variability was 0.7% (club head speed), 4.6% (isometric mid-thigh pull), 3.1% (isometric bench press), 3.8% (cable trunk rotation peak power), 7.0% (cable trunk rotation mean power), 3.1% (isometric leg press), 19.8% (isometric leg press RFD 0-30 ms), 16.5% (isometric leg press RFD 0-50 ms), 13.8% (isometric leg press RFD 0-100 ms), 11.0% (isometric leg press RFD 0-200 ms), 2.1% (CMJ jump height), 2.4% (CMJ positive impulse), 16.1% (CMJ RFD_{dyn} at 0-50 ms), 2.1% (CMJ mean concentric power), 1.9% (CMJ peak concentric power) and 11.6% (CMJ lower limb stiffness).

[Insert Figure 2 and Table 3 approximately here]

381 **DISCUSSION**

382 The main finding of the present study was that maximal club head speed (CHS) was strongly
383 influenced by maximal muscle strength (isometric mid-thigh pull, isometric bench press), RFD and
384 RFD_{dyn} (CMJ, isometric leg press), positive impulse, jump height (CMJ) and muscle power (CMJ,
385 cable trunk rotation) in the present group of highly skilled male golfers. Expanding these
386 observations, the FTG group demonstrated greater isometric strength, leg press RFD, rotational
387 power, CMJ performance (except RFD_{dyn}), and reduced hip joint mobility compared to the STG
388 group.

389 *Anthropometric measures*

390 Maximal CHS was unrelated to any anthropometric measures. In line with this, no significant
391 differences in anthropometry were observed between FTG and STG. Previous studies have reported
392 positive correlations between CHS and body mass, body height and arm length in skilled golfers
393 (Coughlan et al. 2020; Hellström, 2008; Torres-Ronda et al. 2014), nonetheless the present results
394 suggest that such relations may not necessarily exist for a homogenous group of highly skilled
395 golfers. It could be speculated that CHS is related to fat-free body mass as opposed to total body
396 mass, since skeletal muscle mass and maximal strength/power are closely related (Moss et al. 1997;
397 Suchomel et al. 2018). Although fat-free body mass was not assessed in the present study, previous
398 observations have shown a stronger correlation between CHS and fat-free mass than total body
399 mass ($r = 0.43$ vs 0.27 respectively) (Keogh et al., 2009), prompting future studies to examine this
400 potential relationship in highly skilled golfers.

401 *Isometric bench press strength*

402 CHS was strongly correlated ($r = 0.76$) to isometric bench press MVC strength, with FTG being
403 36% stronger than STG. While no previous study has reported isometric bench press strength for
404 golfers, the present observations underscore that physical training to improve bench press strength
405 may have significant importance for golf performance. In support of this notion, the pectoralis
406 major, the anterior deltoid muscles and the triceps have been shown to be highly activated during
407 the golf downswing (Jobe et al. 1986; Pink et al. 1993; Okuda et al. 2002), and previous training
408 studies have reported accompanying increases in dynamic bench press strength and maximal CHS
409 (Alvarez et al. 2012; Doan et al. 2006). Moreover, Gordon et al. (2009) observed a strong
410 correlation ($r = 0.69$) between CHS and peck deck chest muscle strength (8RM) which further
411 underscores the potential importance of ensuring high levels of chest muscle strength for optimizing

CHS performance. It seems plausible that golfers with the ability to produce large forces during isometric bench press testing may be better able to stabilize the shoulder girdle during the rotating motion of the golf swing, which could more effectively transfer force and power from leg muscles to the club head, via the core musculature. It can further be speculated that strength training (e.g., bench press, squats, deadlifts etc.) may improve contractile strength and RFD that in turn may enable a forceful and explosive-type execution of the golf swing. Not only main agonist muscles may benefit from such targeted strength training, as also supporting muscles and antagonist muscles may be stimulated as well.

It is known that the time from force onset to reach peak force in human skeletal muscles is 250 – 400 ms (Aagaard. 2003) and may even be up to 800 ms (Edman. 2003). Given the brief duration of the golf downswing (< 300 ms) (Hume et al. 2005), it is plausible that rapid force capacity (i.e., RFD) remains of stronger importance for maximal CHS than maximal upper body strength (Read et al. 2014; Sheehan et al. 2019). Maximal muscle strength and rapid force capacity are to some extent positively inter-related, thus the ability to generate high RFD (as well as high maximal power) is at least in part influenced by the level of maximal muscle strength (Andersen & Aagaard 2006, Cormie et al. 2011; Zaras et al. 2022). The positive correlation presently observed between CHS, and isometric bench press peak force may therefore not be directly related to superior upper body strength *per se*, but could also arise from stronger golfers potentially demonstrating greater upper body muscle RFD.

Presently, a strong relation was observed between CHS and isometric bench press strength, but to enable detailed training recommendations for golfers, future studies should investigate in more detail (i.e., RFD) the specific contribution of several upper extremity muscle groups to CHS performance in a more elaborate manner.

Lower body strength

A very strong relationship ($r = 0.82$) was observed between CHS, and peak force recorded during isometric mid-thigh pull testing. Further, FTG had 48% greater peak force compared to STG, while no between-group difference was seen for peak force during isometric leg press testing. Previous reports have observed moderately correlations between CHS, and isometric mid-thigh pull peak force in skilled male golfers (Wells et al. 2018). Maximizing CHS requires activation of relevant musculature in a sequential order from the legs and lower body through the upper body muscles and ultimately to the distal club head (Hume et al., 2005; Tinmark et al. 2010). Specifically, the hip extensor muscles in the trailing leg appear to be highly activated in the initial phase of the

444 downswing while, the erector spinae muscles on the lead side are increasingly activated during the
445 acceleration phase (Bechler et al. 1995; Watkins et al. 1996). The present observation that CHS was
446 more strongly related to isometric mid-thigh pull strength compared to isometric leg press strength
447 ($r = 0.82$ vs. $r = 0.32$) may relate to the more golf specific functional-type standing position during
448 the isometric mid-thigh pull test, as it may closer resemble the position and muscular activation
449 pattern involved during the golf swing.

450 *Explosive lower body strength*

451 In the present study strong to very strong correlations were observed between CMJ performance
452 variables and maximal CHS, in line with previous reports (Hellstöm et al. 2008; Oranchuk et al.
453 2018; Torres Ronda et al. 2014; Wells et al. 2018; Wells et al. 2019; Wells et al. 2022). A primary
454 observation in the present study was the very strong positive correlation ($r = 0.76$) between CHS
455 and CMJ positive impulse, validating previous observations (Wells et al. 2018; Wells et al. 2019;
456 Wells et al. 2022), that CMJ impulse relates to CHS performance.

457 In the present study a strong positive association ($r = 0.61$) was observed between CHS and
458 CMJ RFD_{dyn}. Although this relationship was not as strong as with other CMJ parameters, such as
459 peak and mean power, a high RFD_{dyn} during the CMJ likely represents an important neuromuscular
460 property for superior CHS performance in high-level golfers. In support of this notion, a high RFD,
461 particularly during the early contraction phase (0-50 ms) is highly important for maximizing the
462 area under the force-time curve (the force-time integral), which equals the kinetic impulse (mass ·
463 velocity) produced during this time interval (Aagaard et al. 2002, Maffiuletti et al. 2016). Greater
464 RFD_{dyn} during the eccentric CMJ phase will produce proportional increases in total impulse, which
465 in turn will allow the golfer to produce a given braking impulse in shorter time. Consequently, high
466 CMJ RFD_{dyn} will enable the golfer to perform stretch shortening cycle muscle actions more rapidly
467 (i.e., reaching faster speeds) and to reach higher muscle forces, hence allowing to perform a more
468 powerful concentric golf swing phase potentially resulting in higher CHS at ball impact.

469 The observed relationship between CMJ RFD_{dyn} and CHS along with the elevated CMJ power
470 production and tendency ($P = 0.11$) for greater CMJ RFD_{dyn} observed in FTG altogether justify to
471 classify the highly skilled golfers with high CHS as “explosive-type jumpers” characterized by an
472 ability to perform very rapid and powerful lower limb stretch shortening cycle movements.

473 Moreover, in the present study a very strong positive relationship ($r = 0.77$) emerged between
474 maximal CHS and CMJ lower limb stiffness, supporting previous observations by Sheehan et al.
475 (2019). Lower limb stiffness has previously been positively associated with maximal muscle

476 strength (Kubo et al. 2001), RFD (Brughelli & Cronin, 2008), and high CMJ power output (Laffaye
477 et al. 2005). Notably, in the present study all these variables were positively related to maximal
478 CHS. Hence, increasing lower limb stiffness (for example by increasing maximal eccentric muscle
479 strength) may potentially increase CHS in golfers. Notably, maximal eccentric muscle strength has
480 previously been reported to increase in response to heavy-resistance strength training (Aagaard et
481 al. 1996, Andersen et al. 2005).

482 In the present study very strong positive correlations ($r = 0.75$ vs $r = 0.73 - 0.81$) were observed
483 between CHS vs. CMJ jump height, CMJ peak power and CMJ mean power, respectively. In
484 contrast, other studies have observed CMJ power parameters to be stronger correlated to CHS than
485 CMJ jump height (Coughlan et al. 2020; Hellström et al. 2008; Read et al. 2013). These conflicting
486 findings could be related to body mass such that subjects with higher body mass (i.e., higher muscle
487 mass) would likely produce greater force and power for the same jump height compared to subjects
488 with lower mass. In support of CMJ jump height as a valid test modality in highly skilled golfers,
489 research incorporating different training modalities have observed increases in maximal CMJ jump
490 height along with improvements in CHS (Alvarez et al. 2012; Oranchuk et al. 2018). The present
491 results underline that maximal CMJ jump height is a relevant measure which is related to CHS in
492 highly skilled golfers. However, in cases of increased body mass following periods of strength
493 training, CMJ positive impulse and peak power and mean power may likely be more relevant test
494 parameters than maximal CMJ jump height.

495 Furthermore, in the present study isometric leg press RFD (0-30, 0-50, 0-100, 0-200 ms) were
496 more strongly related to CHS than isometric leg press MVC ($r = 0.59-0.70$ vs. $r = 0.32$), altogether
497 indicating that the ability to rapidly generate force in the leg extensor muscles is of importance for
498 maximizing CHS. Notably, it has previously been suggested that the golf swing represents a slow
499 rather than fast stretch shortening cycle activity (Read et al. 2013). In support of this notion, Wells
500 et al. (2018) examined the association between CHS vs. vertical ground reaction forces, impulse
501 and RFD produced during vertical jumping (CMJ, squat jumps, drop-jumps) and isometric mid-
502 thigh pull testing, respectively, in highly skilled golfers, reporting that force and impulse
503 characteristics related to slow stretch shortening cycle movements (i.e. CMJ) were more strongly
504 correlated to maximal CHS than movements involving fast (i.e. squat jumps, drop-jumps) stretch
505 shortening cycle movements. The authors speculated that the association observed between CHS,
506 and slow stretch shortening cycle movements could be a result of skilled golfers utilizing a swing
507 technique where ground reaction forces start to build up during the backswing, as the upper body is

508 still rotating away from the ball. This notion is backed by a recent study by Han et al. (2019)
509 showing that ground reaction forces during the golf swing usually start ≈ 600 ms prior to ball
510 impact, which would include the final part of the backswing phase. Consequently, highly skilled
511 golfers may be able to extend the time available for lower body force production. In contrast, the
512 findings of the present study support the theory that high CHS may also rely on the ability to
513 produce leg ekstensor force rapidly (i.e. high RFD), potentially due to the brief duration of the golf
514 downswing (< 300 ms) (Read et al. 2014) and the even shorter duration of the final acceleration
515 phase of the downswing (defined as the period between the club reaching horizontal position and
516 ball impact), which involves the greatest level of muscle activity (McHardy & Pollard, 2005).

517 *Rotational trunk power*

518 The assessment of rotational trunk power in golfers has been limited to machine based, strength
519 measurements (RM) or medicine ball test protocols (Coughlan et al. 2020; Donahue et al. 2021;
520 Gordon et al. 2009; Keogh et al. 2009; Read et al. 2013; Torres-Ronda et al. 2014). In the present
521 study, cable trunk rotation peak power was found to be greater (+37%) in FTG compared to STG
522 players, and very strongly correlated ($r = 0.81$) to maximal CHS. Keogh et al. (2009) previously
523 reported a strong positive correlation ($r = 0.71$) between estimated 1-RM strength in a golf specific
524 trunk rotation cable test and maximal CHS, and with low handicap golfers being stronger (28%)
525 than higher handicap golfers. It was speculated that the principle of specificity influenced the
526 relationship between CHS and the golf specific trunk rotation cable test, given that the test
527 movement and the golf swing were deemed highly similar in terms of posture, ROM, intended
528 velocity, direction of force application, and kinematic sequence patterns (Keogh et al. 2009). The
529 present results support this notion. However, peak power may be a more valid test outcome with the
530 trunk rotation test than estimated 1-RM strength, as peak power represents the greatest
531 instantaneous exerted power during a single movement performed with the goal of producing
532 maximal velocity at takeoff, release, or impact (Cormie et al. 2011). Since the golf downswing is a
533 quick and powerful motion aiming to produce peak CHS at ball impact, it seems reasonable to
534 expect that the peak power measured during maximal effort cable trunk rotation testing would be
535 highly transferable to the golf swing.

536 *Active range of joint motion*

537 Rotational AROM measures for the trunk and shoulder joints were unrelated to maximal CHS in the
538 present study. Nonetheless, FTG demonstrated reduced rotational AROM for left (-16%) and right
539 (-17%) hips compared to STG, which confirms previous results by Keogh et al. (2009). Likewise,

540 Sheehan et al (2019) observed a negative correlation between lead hip joint internal rotation and
541 maximal CHS and noted that less hip joint range of motion would result in more angular separation
542 between the pelvis and upper torso at the top of the backswing or early in the downswing (termed
543 the 'x-factor' and 'x-factor stretch', respectively) (Keogh et al. 2009; Sheehan et al. 2019). The x-
544 factor stretch is considered to enhance the stretch shortening cycle movement, in turn leading to
545 increased force and impulse production (Hume et al. 2005; Joyce. 2017). Conversely, it has been
546 suggested that greater hip ROM facilitates a longer backswing to hereby enhance the time for force
547 production and hence greater CHS (Keogh et al. 2009). The present group of less flexible highly
548 skilled golfers (FTG) were able to generate superior CHS, which supports the notion that reduced
549 hip joint AROM may yield a positive effect on golf swing kinematics.

550 Notably, a general transferability from the applied lying/sitting AROM tests to the dynamic
551 nature of the golf swing may be questioned.

552 *Golf performance*

553 Although not a primary focus of the present study, a strong correlation ($r = 0.59$) was observed
554 between CHS and competitive golf performance (HCP). Furthermore, FTG demonstrated higher
555 (+11 %) CHS as well as greater HCP than STG, which emphasizes that maximal CHS is an
556 important skill that is closely linked to golf performance.

557 *Study limitations*

558 This study examined young highly skilled male golfers, and care should be taken to extend the
559 findings to other populations such as e.g., female golfers, youth golfers, or senior golfers > 28
560 years. Separate correlational studies should be performed in female golfers as well as more
561 experienced (older) elite and high-level golfers, respectively.

562 Moreover, longitudinal studies in highly skilled golfers are warranted to establish casual effects.
563 Furthermore, future research studies could benefit from incorporating other important elements of
564 golf performance such as driving accuracy, readiness to perform and short game effectiveness.

565 Finally, participants performed several tests within a timeframe of approximately two hours,
566 thus fatigue may have had a confounding influence on individual performance.

567

568 **CONCLUSIONS**

569 Young highly skilled male golfers with high driving CHS demonstrated significantly greater lower
570 and upper body maximal strength, rotational trunk peak power, isometric leg press RFD, CMJ
571 performance and less hip joint flexibility compared to highly skilled golfers characterized by slower
572 CHS. In addition, very strong positive associations were observed between maximal CHS, and
573 isometric mid-thigh pull peak force, isometric leg press RFD, isometric bench press peak force,
574 rotational trunk peak power and CMJ performance, indicating that these force/power parameters
575 should be improved with targeted strength/power training in order to increase maximal CHS in
576 young highly skilled male golfers.

577

578 PRACTICAL IMPLICATIONS

579 Practitioners and coaches in golf should implement the present strength and power tests for
580 monitoring physical capacity in golfers, especially during training periods where the aim is to
581 improve maximal CHS. The present study further suggests that training protocols to effectively
582 improve vertical ground reaction force, maximal and explosive lower body strength and power,
583 maximal upper body strength and rotational trunk power should be implemented for highly skilled
584 golfers, as this would be expected to improve golf performance by increasing CHS and thereby
585 improving maximal driving distance. Notably, maximal CHS was positively correlated with
586 competitive golf performance (HCP), underlining that improving CHS should be considered a key
587 training goal for highly skilled golfers. Additionally, given the strong positive associations observed
588 between CHS and CMJ performance, peak force during the isometric mid-thigh pull, and peak force
589 during isometric bench press, respectively, less strong highly skilled golfers would be
590 recommended to primarily focus on increasing contractile strength through designated resistance
591 training. Interestingly, previous research in non-golfing populations have indicated high-intensity
592 (i.e., heavy-load) strength exercise to represent a more effective training modality to improve
593 athletic performance (e.g., maximal jump, sprint capacity) and increase maximal muscle strength
594 compared to power training using low loading intensities (Cormie et al., 2010). Given the present
595 strong relationship between isometric leg press RFD and CHS, advanced golfers may respond
596 optimally if exposed to explosive-type resistance training, especially when considering golfers with
597 already high levels of maximal muscle strength and muscle mass. Finally, although not examined in
598 the present study, designated strength and conditioning protocols may also reduce the risk of
599 musculoskeletal injury as well as positively influence the physical state and wellbeing of golfers
600 (Brearley et al. 2019; Lauersen et al. 2014).

601

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606

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608 No potential conflict of interest was reported by the authors.

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867 **TABLES**

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Table 1: Characteristics of the total population and descriptive statistics and correlations of FTG and STG golfers

Characteristics	Total population (N = 21) Mean ± SD	Tertile 1 (FTG; N = 7) Mean ± SD	Tertile 2 (STG; N = 7) Mean ± SD	%-diff between FTG and STG	P (two-tailed) FTG vs STG
Age (yr)	21.9 ± 3.9	22.7 ± 3.8	20.7 ± 5.02	9.9%	--
Weight (kg)	80.6 ± 11.9	81.4 ± 12.1	74.9 ± 8.9	8.7%	0.28
Height (cm)	184.6 ± 4.4	183 ± 5.9	184.3 ± 2.8	-0.71%	0.61
Mean arm length (cm)	75.0 ± 2.7	74.4 ± 3.9	74.9 ± 1.6	-0.67%	0.76
Club head speed (km·t ⁻¹)	187.7 ± 9.5	198.5 ± 6.6	178.4 ± 2.6	11.3%	<0.0001**
Golf handicap	+1.1 ± 1.7	+2.1 ± 1.3	+0.09 ± 1.9	--	0.04*

869 *P < 0.05, **P < 0.001, -- = not calculated

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Table 2: Descriptive statistics for MVC strength, rotational trunk power, countermovement jump variables and active range of motion along with correlation coefficients and coefficient of determination (r^2) for relationships with CHS

Parameter	Mean $\pm$ SD	r	r^2	P (two-tailed)
Isometric leg press RFD 0-30 ms (N/s)	10508 $\pm$ 6774	0.59	0.35	0.01*
Isometric leg press RFD 0-50 ms (N/s)	12189 $\pm$ 7864	0.63	0.40	0.006*
Isometric leg press RFD 0-100 ms (N/s)	11887 $\pm$ 6717	0.65	0.42	0.005*
Isometric leg press RFD 0-200 ms (N/s)	10282 $\pm$ 3560	0.70	0.49	0.002*
CMJ jump height (cm)	36.1 $\pm$ 5.6	0.75	0.56	<0.0001**
CMJ positive impulse (N·s)	311.7 $\pm$ 47	0.76	0.57	<0.0001**
CMJ RFD _{dyn} at 0-50 ms (N/s)	4775 $\pm$ 2441	0.61	0.37	0.004*
CMJ mean concentric power (W)	2072 $\pm$ 386.9	0.81	0.66	<0.0001**
CMJ lower limb stiffness (N/m)	5505 $\pm$ 1738	0.77	0.60	<0.0001**
Total rotational AROM right shoulder (°)	154,5 $\pm$ 7,8	-0.17	0.03	0.46
Total rotational AROM left shoulder (°)	156,3 $\pm$ 8,7	-0.27	0.07	0.23
Total rotational AROM right hip (°)	85,1 $\pm$ 14,8	-0.42	0.17	0.06
Total rotational AROM left hip (°)	81,6 $\pm$ 16.2	-0.36	0.13	0.1
Total rotational AROM torso (°)	145.1 $\pm$ 7.7	-0.31	0.1	0.17
Golf handicap	+1.1 $\pm$ 1.7	0.59	0.35	0.005*

*P < 0.05, **P < 0.001, RFD_{dyn} = dynamic rate of force development, RFD = rate of force development = AROM = active range of motion, CMJ = countermovement jump, MVC = maximal voluntary contraction

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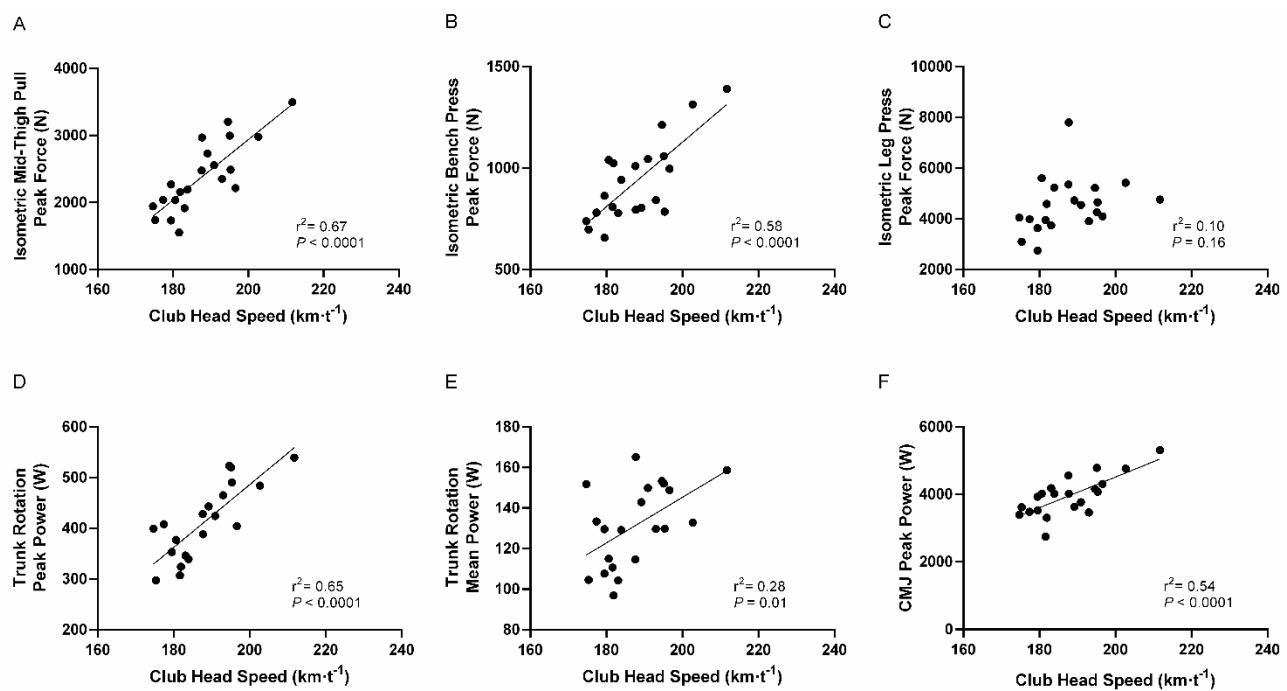
Table 3: MVC strength, rotational trunk power, countermovement jump variables and active range of motion obtained in FTG and STG (Mean $\pm$ SD).

Parameter	FTG (N = 7) Mean $\pm$ SD	STG (N = 7) Mean $\pm$ SD	%-diff between FTG and STG	P (two-tailed)
Isometric leg press RFD 0-30 ms (N/s)	14334 $\pm$ 6791	7026 $\pm$ 5197	104%	0.04*
Isometric leg press RFD 0-50 ms (N/s)	16932 $\pm$ 7803	7850 $\pm$ 5660	116%	0.04*
Isometric leg press RFD 0-100 ms (N/s)	16208 $\pm$ 5701	7562 $\pm$ 5069	114%	0.018*
Isometric leg press RFD 0-200 ms (N/s)	13148 $\pm$ 2421	7173 $\pm$ 2953	83%	0.009*
CMJ jump height (cm)	41 $\pm$ 5.3	32.1 $\pm$ 3.9	28%	0.004*
CMJ positive impulse (N·s)	347 $\pm$ 42.1	275 $\pm$ 25.6	26%	0.002*
CMJ RFD _{dyn} at 0-50 ms (N/s)	6132 $\pm$ 3477	3627 $\pm$ 461	69%	0.11
CMJ mean concentric power (W)	2418 $\pm$ 432	1819 $\pm$ 210	33%	0.006*
CMJ lower limb stiffness (N/m)	7029 $\pm$ 2349	4607 $\pm$ 902	53%	0.04*
Total rotational AROM right shoulder (°)	151.3 $\pm$ 6	154.9 $\pm$ 6.6	-2.3%	0.82
Total rotational AROM left shoulder (°)	151.1 $\pm$ 7.3	159.1 $\pm$ 6.6	-5.03%	0.052
Total rotational AROM right hip (°)	77 $\pm$ 12	93 $\pm$ 5	-17%	0.01*
Total rotational AROM left hip (°)	75 $\pm$ 13	89 $\pm$ 5	-16%	0.01*
Total rotational AROM torso (°)	141 $\pm$ 8	147 $\pm$ 4	-4.08%	0.10

*P < 0.05, **P < 0.001, RFD_{dyn} = dynamic rate of force development, RFD = rate of force development, AROM = active range of motion, CMJ = countermovement jump, MVC = maximal voluntary contraction

876 **FIGURES**

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879 **Figure 1:** Relationships between maximal club head speed and peak force (N) or power (W) produced in isometric mid-
880 thigh pull (A), isometric bench press (B), isometric leg press (C), cable trunk rotation (D, E) and countermovement
881 jump (F).

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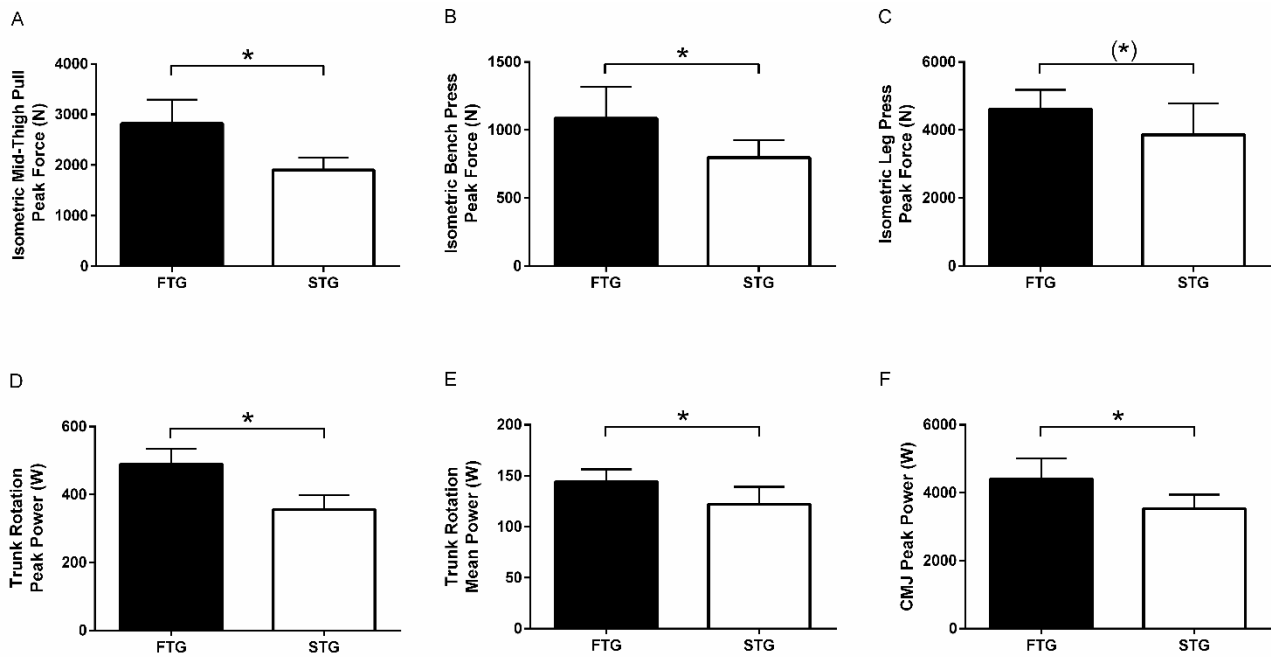


Figure 2: Comparison between golf players with the fastest maximal club head speed (FTG) and players with the lowest maximal club head speed (STG). Differences between FTG and STG in peak force produced in isometric mid-thigh pull (A), isometric bench press (B), isometric leg press (C), and differences in power in cable trunk rotation (D, E) and countermovement jumping (F): *P < 0.05.